

# ROHAN SHARMA

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## Experience

**Darwin Box** · HR & Payroll SaaS · Data & ML Engineer (Contract) Nov 2023 – Jan 2024

- Designed automated ETL pipelines in Python (Pandas, NumPy) and SQL across three data sources, reducing weekly reporting from 8+ hours to 15 minutes.
- Developed statistical trend-detection models for predictive maintenance, enabling earlier failure detection and supporting maintenance planning.
- Identified seven operational bottlenecks through workflow analysis, leading to the implementation of five process improvements.

**Rdr.inc** · Risk Management & Data Analytics · Data Analyst (Contract) May 2022 – Dec 2022

- Analyzed 120K+ records using Python and SQL to identify high-risk patterns that were previously missed in manual review workflows.
- Built predictive risk classification models in scikit-learn, achieving 82% validation accuracy on held-out data.
- Automated data validation and QA checks, reducing manual review effort by 35%.

## Projects

**InkGuard – Signature Forgery Detection System** (*GitHub*) – TensorFlow, FastAPI, Siamese, VAE, SQLite, JWT, Vanilla JS

- Architected a full-stack signature forgery detection system using React, FastAPI, JWT authentication, and an end-to-end machine learning pipeline.
- Developed a hybrid VAE–Siamese Network for signature verification, encompassing data preprocessing, pair generation, model training, inference serving, and model versioning.

**Physics-Informed Neural Networks for 2D Turbulence** (*GitHub*) – ML, Navier-Stokes, Computational Physics

- Implemented a 6-layer Physics-Informed Neural Network (83K parameters) for solving 2D Navier-Stokes equations without simulation labels.
- Achieved PDE residual of  $4.8 \times 10^{-3}$  and BC loss of  $4.4 \times 10^{-3}$ , validated against Ghia et al. (1982).

**Feel CV – Intelligent Resume Parser** (*GitHub*) – Python, LLM APIs (grok), GenAI, React, Browser Extension

- Built a production Chrome extension using LLM APIs to extract structured candidate data from unstructured resume PDFs and auto-populate job application forms, reducing manual data entry by 67%.
- Designed prompt engineering pipeline handling variable resume formats; entity extraction across 12 fields with JSON schema validation before DOM injection.

**Blue Quants – Quantitative Analytics** (*Demo*) – Render, LSTM, ARIMA, NLP Sentiment, REST APIs, SQL

- Built a real-time market intelligence platform integrating prices, volume, technical indicators, and news sentiment with LSTM, ARIMA, and NLP models.
- Reduced trade review time by 2× through automated signal summarization and evidence-based market insights.

## Skills

**Languages:** Python, SQL, JavaScript, Typescript

**ML/AI:** TensorFlow, PyTorch, scikit-learn, LSTM, ARIMA, PINNs, LLM(Claude) APIs

**Data:** Pandas, NumPy, PostgreSQL, MySQL, ETL, Databricks

**Cloud & Tools:** AWS (EC2, S3), GCP (BigQuery), Git, Tableau, Power BI, React

## Education

**National Institute of Technology Sikkim** Expected Jun 2027

M.Tech in Artificial Intelligence & Machine Learning

**Siliguri Institute of Technology**

B.Tech in Computer Science & Engineering

Jul 2024

GPA: 9.0/10

## Certifications & Awards

- AWS Certified Solutions Architect – Advanced Networking
- Databricks AI Application – Evaluation, Governance, Deployment and Monitoring
- CS50 – Harvard University – Data Science using ML (40+ Hours)
- Intel Ideathon & INNOVATEathon 2023 – 1st Runner-up
- LeetCode: 356 problems solved — Top 25th Percentile